

# AURA TRAVEL - Master Production Roadmap

## 16-Week Sprint Plan to Production-Ready System

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### PROJECT OVERVIEW

**Team:** 4 developers (100 hours/week total capacity)  
**Timeline:** 16 weeks (1 semester)  
**Goal:** Full production-ready AI-powered travel platform  
**Budget:** ₹500 (domain only)  
**Tech Stack:** MERN + Leaflet.js + compromise.js

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### SUCCESS CRITERIA

- By Week 16, you will have:**  Fully functional website (frontend + backend)
-  110+ real curated travel entities in MongoDB
  -  Working AI Experience Weaver (NLP → Itinerary)
  -  Working AI Travel Sync (dynamic replanning)
  -  User authentication & admin panel
  -  Mock booking system (flights/hotels)
  -  Interactive maps with routes
  -  Mobile-responsive design
  -  Production-ready code quality
  -  Complete documentation
  -  Demo-ready for presentation
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### 16-WEEK SPRINT BREAKDOWN

#### PHASE 0: FOUNDATION (Week 1)

**Goal:** Development environment ready, team aligned, tech stack validated

#### Week 1: Setup & Architecture Planning

**Team Activity (Everyone):**

- ☐ Development environment setup
- ☐ Git repository created & configured

- ☐ Project folder structure established
- ☐ Design system & UI mockups finalized
- ☐ Database schema designed
- ☐ API contract defined

#### **Pair 1 (Frontend Focus):**

- ☐ React app initialized with Vite/CRA
- ☐ Install dependencies (React Router, Material-UI/Tailwind)
- ☐ Create basic component structure
- ☐ Setup ESLint & Prettier

#### **Pair 2 (Backend Focus):**

- ☐ Node.js + Express server initialized
- ☐ MongoDB Atlas account created (free tier)
- ☐ Basic CRUD API for testing
- ☐ CORS & middleware configured

#### **Deliverables:**

- Working "Hello World" full-stack app
- Team Git workflow established (branching strategy)
- Weekly sprint schedule agreed

**Time Allocation:** 25 hours/person

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### **PHASE 1: CORE INFRASTRUCTURE (Weeks 2-4)**

**Goal:** User system, database, and basic UI operational

#### **Week 2: User Authentication System**

##### **Pair 1 (Frontend - Auth UI):**

- ☐ Login page design & implementation
- ☐ Registration page with validation
- ☐ Protected routes setup
- ☐ User profile page
- ☐ JWT token storage & refresh logic

##### **Pair 2 (Backend - Auth API):**

- ☐ User schema (MongoDB model)
- ☐ POST /api/auth/register endpoint
- ☐ POST /api/auth/login endpoint
- ☐ JWT generation & verification middleware
- ☐ Password hashing (bcrypt)
- ☐ Input validation & error handling

**Deliverables:**

- Users can register and login
- JWT-based authentication working
- Protected routes functional

**Time:** 25 hours/person

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### **Week 3: Database Design & Initial Data**

**Pair 1 (Data Curation Team):**

- ☐ Create MongoDB schemas (Activities, Hotels, Restaurants)
- ☐ Define experiential tag vocabulary (95 tags)
- ☐ Curate 20 activities for Goa (pilot)
- ☐ Assign tags manually (8-12 per entity)
- ☐ Add photos/descriptions

**Pair 2 (Admin Panel - Basic):**

- ☐ Admin login page
- ☐ Dashboard layout
- ☐ CRUD interface for activities
- ☐ Form for adding new entities
- ☐ List view with edit/delete

**Deliverables:**

- 20+ real entities in database (Goa pilot)
- Basic admin panel operational
- All entities have experiential tags

**Time:** 25 hours/person

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## Week 4: Core UI Components & Navigation

### Pair 1 (Frontend - UI Library):

- ☐ Homepage with hero section
- ☐ Navigation bar (responsive)
- ☐ Footer component
- ☐ Card components for activities/hotels
- ☐ Search bar component (visual only)
- ☐ Mobile responsive breakpoints

### Pair 2 (Backend - Search API Foundation):

- ☐ GET /api/activities endpoint
- ☐ GET /api/hotels endpoint
- ☐ GET /api/restaurants endpoint
- ☐ Filtering logic (by destination, tags)
- ☐ Sorting & pagination

### Deliverables:

- Professional-looking homepage
- Browse/search interface (basic)
- API endpoints returning data

**Time:** 25 hours/person

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## PHASE 2: AI EXPERIENCE WEAVER (Weeks 5-8)

**Goal:** Core innovation #1 - NLP-based itinerary generation working

### Week 5: NLP Tag Extraction Module

#### Pair 1 (NLP Implementation):

- ☐ Install compromise.js
- ☐ Create tag extraction function
- ☐ Map adjectives → experiential tags
- ☐ Extract location entities (Goa, Manali, mountains)
- ☐ Extract temporal info (3 days, weekend)

- ☐ Unit tests for NLP module

### **Pair 2 (Conversational UI):**

- ☐ Conversational prompt component
- ☐ "What kind of experience..." input field
- ☐ Real-time tag preview (show extracted tags)
- ☐ Example queries for users
- ☐ Loading states & animations

### **Deliverables:**

- NLP extracts tags with ~88% accuracy
- User can type natural language queries
- System shows extracted tags

**Time:** 25 hours/person

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## **Week 6: Knowledge Graph Query Engine**

### **Pair 1 (Backend - Matching Algorithm):**

- ☐ POST /api/weaver/generate endpoint
- ☐ MongoDB query builder (tag matching)
- ☐ Ranking algorithm implementation
  - $\text{Score} = (\text{tag\_overlap} \times 3) + (\text{rating} \times 2) + \text{budget}$
- ☐ Multi-destination search
- ☐ Response formatting (JSON itinerary)

### **Pair 2 (Frontend - Results Display):**

- ☐ Itinerary results page design
- ☐ Timeline-based layout (day-by-day)
- ☐ Activity cards with images & tags
- ☐ Map integration (basic markers)
- ☐ "Book This Trip" button (placeholder)

### **Deliverables:**

- Query "peaceful mountains 3 days" → Returns Manali itinerary
- Results displayed beautifully

- All tags matched correctly

**Time:** 25 hours/person

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## **Week 7: Itinerary Composition Algorithm**

### **Pair 1 (Backend - Composer):**

- ☐ Day segmentation logic
- ☐ Time-of-day optimization
  - Morning: Cultural/active
  - Afternoon: Leisure/dining
  - Evening: Entertainment
- ☐ Geographical clustering (minimize travel)
- ☐ Activity diversity check (no repetition)
- ☐ Accommodation selection & placement

### **Pair 2 (Frontend - Enhanced Display):**

- ☐ Day-by-day expandable sections
- ☐ Time slots visualization
- ☐ Activity details modal
- ☐ Print/export itinerary (PDF)
- ☐ Share functionality (copy link)

### **Deliverables:**

- Complete 3-day itineraries generated
- Logical flow (morning → afternoon → evening)
- Hotels selected optimally

**Time:** 25 hours/person

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## **Week 8: Data Expansion & Testing**

### **Pair 1 (Data Curation - Scale Up):**

- ☐ Complete Goa dataset (60+ activities total)
- ☐ Start Manali dataset (30+ activities)

- ☐ Add 20+ hotels across destinations
- ☐ Add 15+ restaurants
- ☐ Quality check all tags

**Pair 2 (Testing & Refinement):**

- ☐ Test 20+ different queries
- ☐ Fix edge cases (no results, single activity)
- ☐ Performance optimization (query speed)
- ☐ Error handling improvements
- ☐ User feedback collection

**Deliverables:**

- 110+ entities in database
- AI Experience Weaver working reliably
- Response time < 4.5 seconds

**Time:** 25 hours/person

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**PHASE 3: AI TRAVEL SYNC (Weeks 9-11)**

**Goal:** Core innovation #2 - Dynamic replanning operational

**Week 9: Context Monitoring System**

**Pair 1 (Backend - Live Trip State):**

- ☐ Trip schema (active itineraries)
- ☐ POST /api/trips/start endpoint
- ☐ GET /api/trips/current (get active trip)
- ☐ Activity status tracking (completed/upcoming)
- ☐ Context change detection logic

**Pair 2 (Frontend - Live Trip UI):**

- ☐ "Live Trip" view component
- ☐ Current activity highlight
- ☐ Next activities timeline
- ☐ Progress indicators
- ☐ "Mark as Complete" buttons

**Deliverables:**

- Users can start a "live trip"
- Current itinerary displayed clearly
- Activity tracking functional

**Time:** 25 hours/person

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**Week 10: Constraint Detection & Alternative Search****Pair 1 (Backend - Sync Engine):**

- ☐ POST /api/sync/check-conflicts endpoint
- ☐ Rule-based constraint detection
  - Rain + outdoor → conflict
  - Overcrowding + venue → conflict
- ☐ Alternative search query builder
- ☐ Ranking alternatives (preserve intent)
- ☐ Suggestion formatting

**Pair 2 (Frontend - Simulation Controls):**

- ☐ Developer control panel (for demo)
- ☐ "Simulate Rain" button
- ☐ "Simulate Overcrowding" button
- ☐ "Simulate User Tired" button
- ☐ Context visualization

**Deliverables:**

- Conflict detection working
- System finds suitable alternatives
- Simulation panel functional

**Time:** 25 hours/person

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**Week 11: Proactive Suggestions & UI Polish****Pair 1 (Backend - Finalization):**



- ☐ Response time optimization (< 1.5s)
- ☐ Multiple alternatives (top 3)
- ☐ Reasoning text generation
- ☐ Accept/decline alternative logic
- ☐ Itinerary update on acceptance

**Pair 2 (Frontend - Notification System):**

- ☐ Suggestion notification component
- ☐ Slide-in animation
- ☐ Problem description + solution
- ☐ Accept/Decline/See More buttons
- ☐ Auto-dismiss after 30 seconds

**Deliverables:**

- AI Travel Sync fully operational
- Suggestions appear proactively
- User can accept/decline easily

**Time:** 25 hours/person

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**PHASE 4: BOOKING SYSTEM & MAPS (Weeks 12-13)**

**Goal:** Mock booking flow and interactive maps

**Week 12: Mock Booking Engine**

**Pair 1 (Backend - Mock Data):**

- ☐ Create flights.json (50+ flights)
- ☐ Create hotels.json (30+ hotels)
- ☐ GET /api/flights endpoint
- ☐ GET /api/hotels endpoint
- ☐ Search/filter logic
- ☐ Booking confirmation (mock)

**Pair 2 (Frontend - Booking UI):**

- ☐ Flight search page
- ☐ Flight results display

- ☐ Hotel search page
- ☐ Hotel results with images
- ☐ Booking summary page
- ☐ Confirmation page (mock)

**Deliverables:**

- Users can "book" flights and hotels
- Professional booking flow
- Data looks realistic

**Time:** 25 hours/person

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## **Week 13: Maps Integration (Leaflet.js)**

### **Pair 1 (Maps - Core Features):**

- ☐ Install Leaflet.js + React-Leaflet
- ☐ Initialize map component
- ☐ Add activity markers (pins)
- ☐ Custom marker icons (activity types)
- ☐ Popup on marker click (activity info)
- ☐ Map centering & zoom

### **Pair 2 (Maps - Routes & Interactions):**

- ☐ Draw routes between activities
- ☐ Calculate distances (Leaflet Routing Machine)
- ☐ Show travel time estimates
- ☐ Cluster markers for dense areas
- ☐ Mobile-friendly map controls

**Deliverables:**

- Interactive map showing all activities
- Routes drawn between locations
- Professional appearance

**Time:** 25 hours/person

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## PHASE 5: POLISH & PRODUCTION PREP (Weeks 14-15)

**Goal:** Professional quality, bug-free, optimized

### Week 14: UI/UX Polish & Mobile Optimization

#### Pair 1 (Frontend - Refinement):

- ☐ Consistent color scheme & typography
- ☐ Loading skeletons for all pages
- ☐ Empty states (no results, no trips)
- ☐ Error boundaries (React error handling)
- ☐ Animations & transitions
- ☐ Accessibility improvements (ARIA labels)

#### Pair 2 (Responsive Design):

- ☐ Mobile breakpoints (< 768px)
- ☐ Tablet breakpoints (768-1024px)
- ☐ Touch-friendly buttons (min 44px)
- ☐ Hamburger menu for mobile
- ☐ Optimize images for mobile
- ☐ Test on real devices

#### Deliverables:

- Professional, polished UI
- Perfect mobile experience
- No visual bugs

**Time:** 25 hours/person

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### Week 15: Testing, Bug Fixes, Performance

#### Pair 1 (Quality Assurance):

- ☐ Write test cases (unit tests)
- ☐ Integration testing (API → Frontend)
- ☐ User acceptance testing (UAT)
- ☐ Cross-browser testing (Chrome, Firefox, Safari)
- ☐ Fix all critical bugs

☐ Performance profiling

**Pair 2 (Optimization):**

- ☐ Code splitting (lazy loading)
- ☐ Image optimization (compression, lazy load)
- ☐ Database query optimization
- ☐ Caching strategy (Redis if needed)
- ☐ Bundle size reduction
- ☐ Lighthouse score > 90

**Deliverables:**

- Zero critical bugs
- Fast load times (< 3s)
- Smooth user experience

**Time:** 25 hours/person

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**PHASE 6: DEPLOYMENT & DOCUMENTATION (Week 16)**

**Goal:** Production deployment, documentation, demo preparation

**Week 16: Launch Week**

**Pair 1 (Deployment):**

- ☐ Frontend deployment (Vercel)
- ☐ Backend deployment (Render/Railway)
- ☐ MongoDB Atlas configuration (production)
- ☐ Domain setup (₹500 budget)
- ☐ SSL certificate (free via host)
- ☐ Environment variables configured
- ☐ Health checks & monitoring

**Pair 2 (Documentation):**

- ☐ README.md (GitHub)
- ☐ User manual (PDF)
- ☐ API documentation (Postman collection)
- ☐ Admin guide

- ☐ Setup instructions
- ☐ Demo script for presentation
- ☐ Video walkthrough (5 min)

**Deliverables:**

- Live website accessible online
- Complete documentation
- Demo-ready presentation

**Time:** 25 hours/person

 **WEEKLY COMMITMENT TRACKER**

| Week | Pair 1 Focus       | Pair 2 Focus      | Total Hours | Deliverable            |
|------|--------------------|-------------------|-------------|------------------------|
| 1    | Frontend Setup     | Backend Setup     | 100h        | Dev environment ready  |
| 2    | Auth UI            | Auth API          | 100h        | Login/Register working |
| 3    | Data Curation      | Admin Panel       | 100h        | 20+ entities in DB     |
| 4    | Core UI            | Search API        | 100h        | Homepage complete      |
| 5    | NLP Module         | Conversational UI | 100h        | Tag extraction working |
| 6    | Matching Algorithm | Results Display   | 100h        | Query → Results        |
| 7    | Composer Algorithm | Enhanced Display  | 100h        | Complete itineraries   |
| 8    | Data Expansion     | Testing & QA      | 100h        | 110+ entities          |
| 9    | Live Trip State    | Live Trip UI      | 100h        | Trip tracking          |
| 10   | Conflict Detection | Simulation Panel  | 100h        | AI Sync foundation     |
| 11   | Suggestions        | Notifications     | 100h        | AI Sync complete       |
| 12   | Mock Booking API   | Booking UI        | 100h        | Booking flow           |
| 13   | Maps Core          | Maps Routes       | 100h        | Interactive maps       |
| 14   | UI Polish          | Responsive Design | 100h        | Mobile-ready           |

| Week | Pair 1 Focus | Pair 2 Focus  | Total Hours | Deliverable        |
|------|--------------|---------------|-------------|--------------------|
| 15   | Testing      | Optimization  | 100h        | Production quality |
| 16   | Deployment   | Documentation | 100h        | LAUNCH!            |

**Total:** 1,600 hours (400 hours/person)



MILESTONE CHECKPOINTS

End of Month 1 (Week 4):

- ☒ Users can login
- ☒ 20+ activities in database
- ☒ Basic homepage functional

**Demo:** Show user authentication & browse activities

End of Month 2 (Week 8):

- ☒ AI Experience Weaver operational
- ☒ 110+ entities curated
- ☒ Itineraries generated from natural language

**Demo:** "I want a peaceful beach trip" → Full 3-day Goa itinerary

End of Month 3 (Week 12):

- ☒ AI Travel Sync working
- ☒ Mock booking flow complete
- ☒ Maps integrated

**Demo:** Simulate rain → System suggests indoor alternative

End of Month 4 (Week 16):

- ☒ Production-ready system
- ☒ Deployed online
- ☒ Documentation complete

**Demo:** Full end-to-end user journey

## **RISK MITIGATION**

### **Risk 1: Data Curation Taking Too Long**

#### **Mitigation:**

- Start with semi-real data (real names, simplified descriptions)
- Use web scraping tools (BeautifulSoup) to speed up
- Divide destinations among team (each person = 1 destination)

### **Risk 2: NLP Accuracy Below 88%**

#### **Mitigation:**

- Focus on simple queries first
- Build comprehensive tag dictionary
- Add Gemini API fallback for complex queries (Week 10+)

### **Risk 3: Running Behind Schedule**

#### **Mitigation:**

- Daily 15-min standups (track blockers)
- Move Priority 2 features to post-launch
- Reduce admin panel to basic version

### **Risk 4: Technical Blockers**

#### **Mitigation:**

- Stack Overflow / Documentation first
- Team pair programming (solve together)
- Prof. Radhika Malpani as escalation point

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## **DEFINITION OF DONE (WEEK 16)**

### **Production-Ready Checklist:**

- ☐ All Priority 1 features implemented & tested
- ☐ Zero critical bugs, < 5 minor bugs

- ☐ Mobile responsive (tested on 3+ devices)
- ☐ Page load time < 3 seconds
- ☐ 110+ real entities with tags in database
- ☐ User authentication secure (JWT, bcrypt)
- ☐ Admin panel functional
- ☐ Documentation complete
- ☐ Deployed online with custom domain
- ☐ Demo video recorded
- ☐ Code on GitHub with README

**When all checked → PRODUCTION READY! 🚀**

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## 📞 TEAM COORDINATION

### Daily Standup (15 min):

- What did I complete yesterday?
- What will I do today?
- Any blockers?

### Weekly Sprint Review (Friday, 1 hour):

- Demo week's work
- Review vs plan
- Plan next week

### Tools:

- **Git:** Feature branches, PRs, code reviews
  - **Trello/Jira:** Task tracking
  - **Slack/Discord:** Team communication
  - **Figma:** UI/UX designs
  - **Postman:** API testing
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## **NEXT STEPS - START NOW!**

### **TODAY (Day 1):**

1. Create GitHub repository
2. Initialize React app (Pair 1)
3. Initialize Express app (Pair 2)
4. Setup MongoDB Atlas account
5. Create project folder structure

### **THIS WEEK (Week 1):**

- Follow Week 1 checklist above
- Setup development environment
- Design database schema
- Create UI mockups

**You're ready to start coding! Check the other artifacts for detailed technical specs.**  